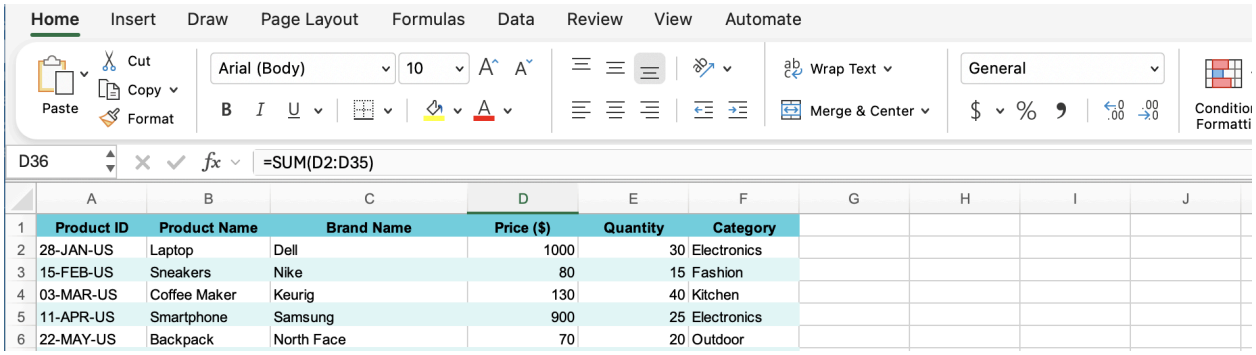
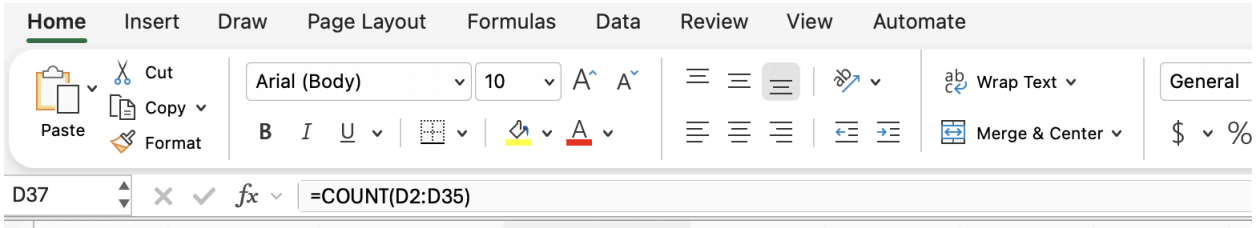
1. Sum, Count, and Average

Perform the following calculations using Excel functions:

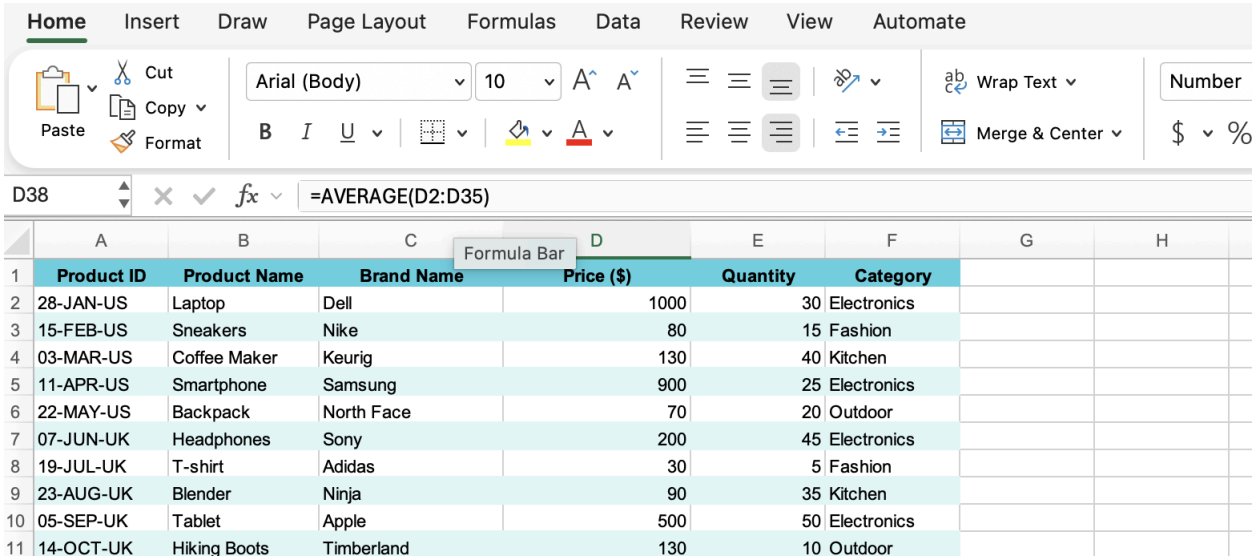
- Calculate the total price of all products in the dataset.
- Determine how many products exist in the dataset.
- Calculate the average price of the products.



	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category				
2	28-JAN-US	Laptop	Dell	1000	30	Electronics				
3	15-FEB-US	Sneakers	Nike	80	15	Fashion				
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen				
5	11-APR-US	Smartphone	Samsung	900	25	Electronics				
6	22-MAY-US	Backpack	North Face	70	20	Outdoor				



	A	B	C	D	E	F	G	H
1	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category		
2	28-JAN-US	Laptop	Dell	1000	30	Electronics		
3	15-FEB-US	Sneakers	Nike	80	15	Fashion		
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen		
5	11-APR-US	Smartphone	Samsung	900	25	Electronics		
6	22-MAY-US	Backpack	North Face	70	20	Outdoor		
7	07-JUN-UK	Headphones	Sony	200	45	Electronics		
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion		
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen		
10	05-SEP-UK	Tablet	Apple	500	50	Electronics		
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor		

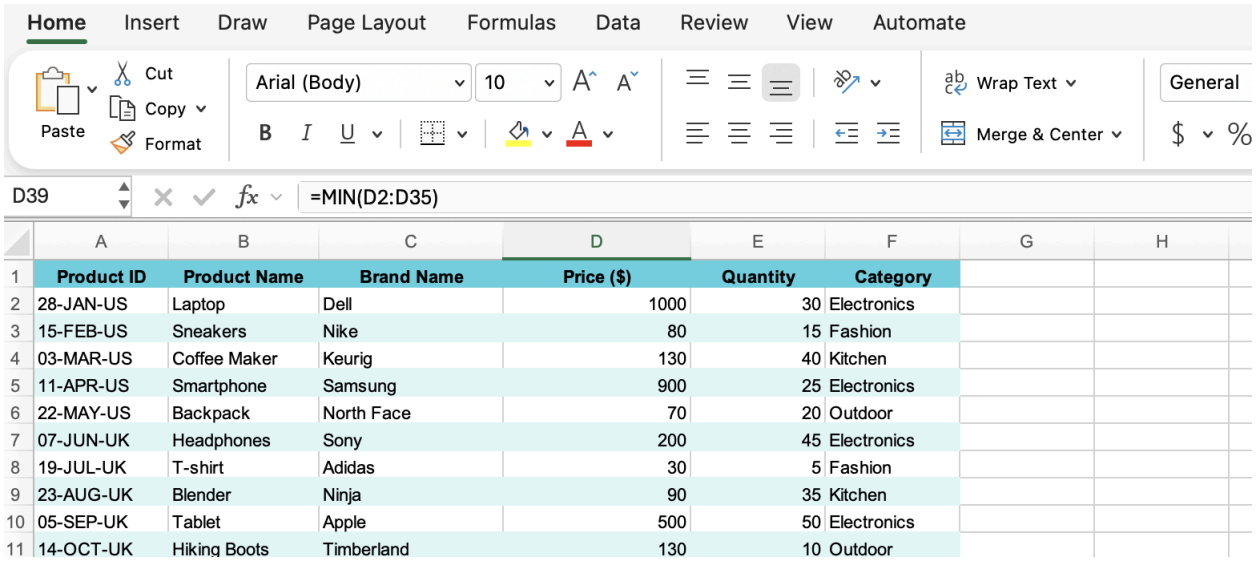


	A	B	C	D	E	F	G	H
1	**Product ID**	**Product Name**	**Brand Name**	**Price ($)**	**Quantity**	**Category**		
2	28-JAN-US	Laptop	Dell	1000	30	Electronics		
3	15-FEB-US	Sneakers	Nike	80	15	Fashion		
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen		
5	11-APR-US	Smartphone	Samsung	900	25	Electronics		
6	22-MAY-US	Backpack	North Face	70	20	Outdoor		
7	07-JUN-UK	Headphones	Sony	200	45	Electronics		
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion		
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen		
10	05-SEP-UK	Tablet	Apple	500	50	Electronics		
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor		

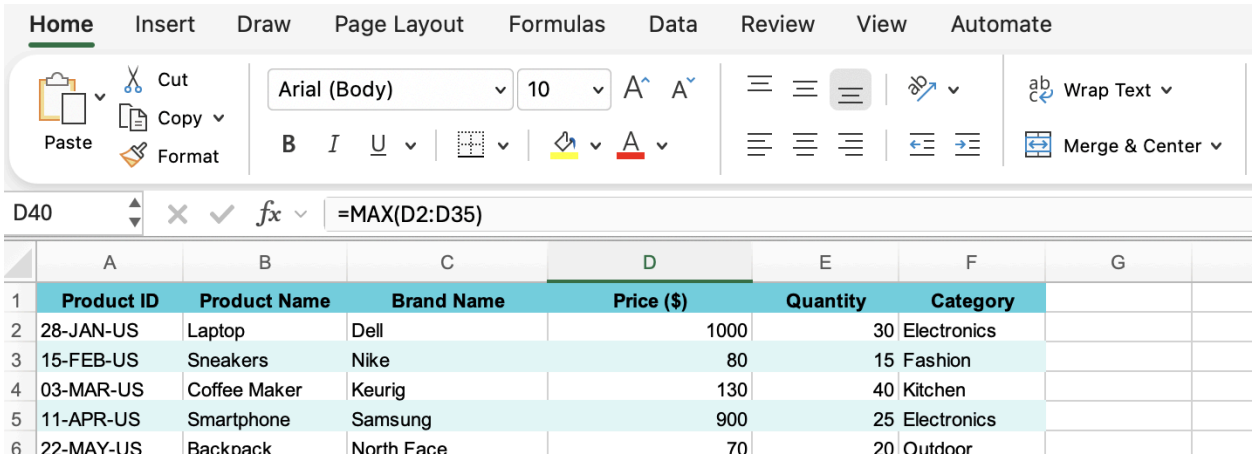
2. Minimum and Maximum Values

Using Excel functions:

- Identify the minimum price among all products.
- Identify the maximum price among all products.



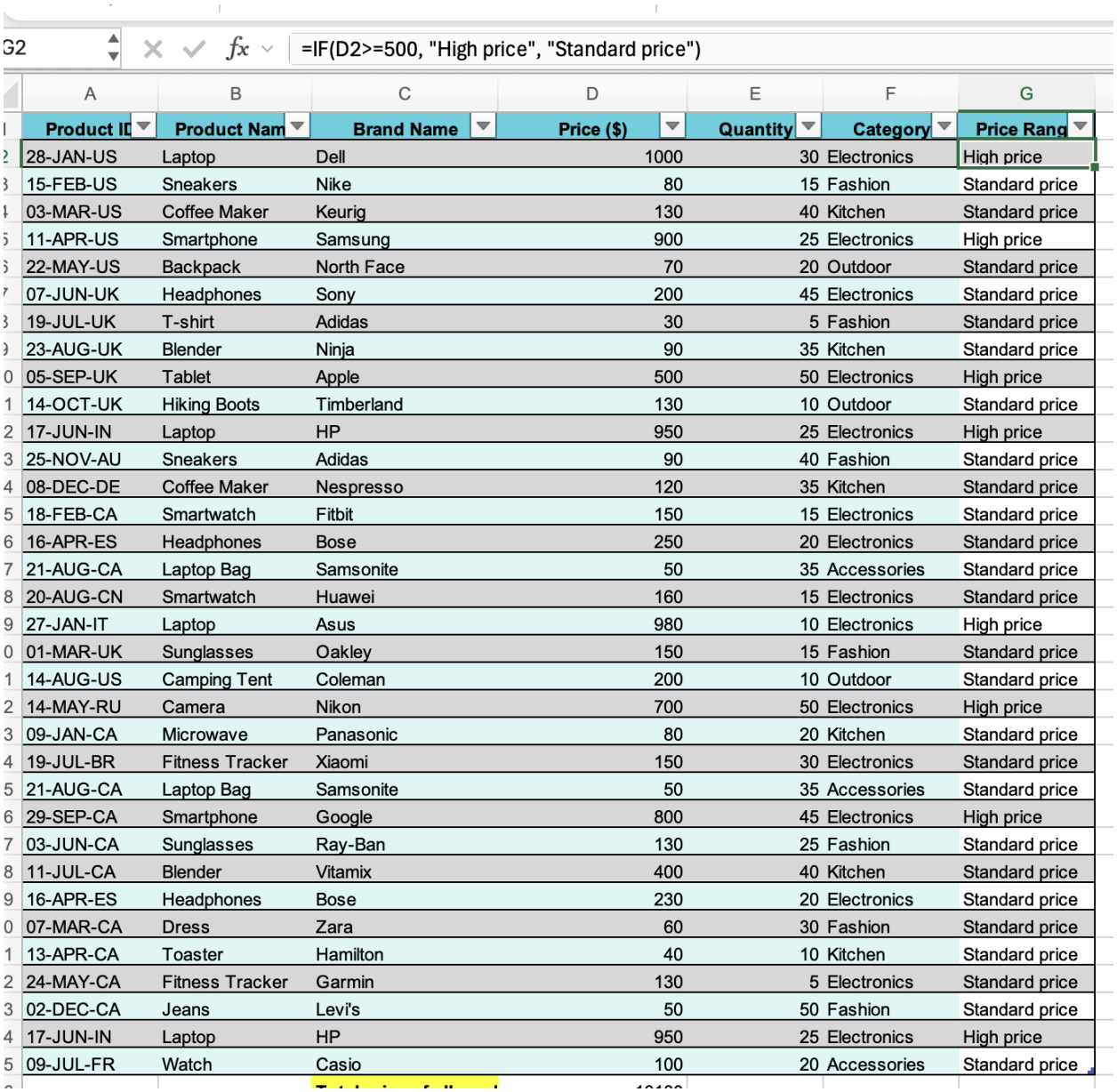
	A	B	C	D	E	F	G	H
1	**Product ID**	**Product Name**	**Brand Name**	**Price ($)**	**Quantity**	**Category**		
2	28-JAN-US	Laptop	Dell	1000	30	Electronics		
3	15-FEB-US	Sneakers	Nike	80	15	Fashion		
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen		
5	11-APR-US	Smartphone	Samsung	900	25	Electronics		
6	22-MAY-US	Backpack	North Face	70	20	Outdoor		
7	07-JUN-UK	Headphones	Sony	200	45	Electronics		
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion		
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen		
10	05-SEP-UK	Tablet	Apple	500	50	Electronics		
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor		



	A	B	C	D	E	F	G
1	**Product ID**	**Product Name**	**Brand Name**	**Price ($)**	**Quantity**	**Category**	
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	

3. Logical Function – IF

- Create a new column named Price Range.
- Use the IF function to categorize products based on their price:
  - If Price ≥ $500 → classify as High Price
  - If Price < $500 → classify as Standard Price



=IF(D2>=500, "High price", "Standard price")

	A	B	C	D	E	F	G
1	Product ID	Product Nam	Brand Name	Price ($)	Quantity	Category	Price Rang
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High price
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard price
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard price
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High price
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard price
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard price
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard price
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard price
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High price
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard price
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High price
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard price
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard price
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard price
16	16-APR-ES	Headphones	Bose	250	20	Electronics	Standard price
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard price
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard price
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	High price
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard price
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard price
22	14-MAY-RU	Camera	Nikon	700	50	Electronics	High price
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard price
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard price
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard price
26	29-SEP-CA	Smartphone	Google	800	45	Electronics	High price
27	03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard price
28	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	Standard price
29	16-APR-ES	Headphones	Bose	230	20	Electronics	Standard price
30	07-MAR-CA	Dress	Zara	60	30	Fashion	Standard price
31	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	Standard price
32	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	Standard price
33	02-DEC-CA	Jeans	Levi's	50	50	Fashion	Standard price
34	17-JUN-IN	Laptop	HP	950	25	Electronics	High price
35	09-JUL-FR	Watch	Casio	100	20	Accessories	Standard price

4. Conditional Functions – SUMIF and COUNTIF

Perform the following conditional calculations:

● Use the SUMIF function to calculate the total price of products in the Electronics category.

● Use the COUNTIF function to determine the number of products with a price less than $100.

D41 =SUMIF(Table1[Category],"Electronics", Table1[Price ($)])

	A	B	C	D	E	F	G
1	Product ID	Product Nam	Brand Name	Price ($)	Quantity	Category	Price Rang
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High price
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard price
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard price
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High price
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard price
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard price
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard price
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard price

D42 =COUNTIF(Table1[Price ($)], "<100")

	A	B	C	D	E	F	G
1	Product ID	Product Nam	Brand Name	Price ($)	Quantity	Category	Price Rang
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High price
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard price
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard price
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High price
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard price
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard price
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard price
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard price

5. Text Formatting – LEFT, RIGHT, MID

● Use text functions to extract information from the Product ID column.

● Create the following new columns:

○ Create a new column named Day with the first 2 characters of each 'Product ID' using the LEFT function.

○ Create a new column named Country Code by extracting the last 2 characters from the 'Product ID' column using the RIGHT function.

○ Create a new column named Month by extracting the 4th to 6th characters from the 'Product ID' column using the MID function.

H2 =LEFT([@[Product ID]],2)

	A	B	C	D	E	F	G	
1	Product ID	Product Nam	Brand Name	Price ($)	Quantity	Category	Price Rang	
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High price	28
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard price	15
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard price	03
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High price	11
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard price	22
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard price	07
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard price	19
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard price	23
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High price	05

I2 =RIGHT(A2,2)

	A	B	C	D	E	F	G
1	Product ID	Product Nam	Brand Name	Price ($)	Quantity	Category	Price Rang
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High price
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard price
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard price
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High price
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard price
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard price
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard price
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard price
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High price

J2 =MID([@[Product ID]],4,3)

	A	B	C	D	E	F	G
1	Product ID	Product Nam	Brand Name	Price ($)	Quantity	Category	Price Rang
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High price
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard price
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard price
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High price
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard price
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard price
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard price
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard price
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High price